

ROHAN JOSHI

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SUMMARY

4+ years of industry experience building production ML and cloud systems, with hands-on experience mentoring engineers, leading code reviews, and explaining complex technical concepts.

EDUCATION

Northeastern University

Seattle, WA

M.S. in Computer Science, GPA: 4.0/4.0

Sept. 2025 – May 2027 (Expected)

Key Courses: Algorithms, Natural Language Processing, Reinforcement Learning, Scalable Distributed Systems

Birla Institute of Technology and Science (BITS Pilani)

Goa, India

B.E. in Computer Science

Aug. 2017 – May 2021

Key Courses: Neural Networks, Artificial Intelligence, Machine Learning, Data Structures, Operating Systems

EXPERIENCE

Cloudwick Technologies

Bengaluru, India

Software Engineer II

May 2024 – Jul. 2025

- Mentored 3 junior engineers and led a 4-person intern team through code reviews, technical walkthroughs, and establishing cloud-native development standards
- Architected *Amorphic AI*, a production RAG framework with semantic search over large-scale document corpora, using multi-agent orchestration (LangChain + AWS Bedrock) with semantic retrieval and metadata filtering over pgvector
- Built an agentic SQL chatbot using AWS Bedrock with an error recovery loop that analyzed failed queries, retrieved additional schema context, and regenerated SQL, significantly reducing query resolution time; implemented post-retrieval access control for row-level security across multi-tenant databases; earned **Cloudwick Spotlight Award**
- Designed automated data pipelines using AWS Glue, exposing ETL workflows as REST APIs via API Gateway with S3 event integration for on-demand and event-driven execution, supporting configurable LLM-powered nodes for document summarization and structured insights extraction

Software Engineer I

Jun. 2021 – May 2024

- Built scalable Intelligent Document Processing system handling large volumes of OCR documents with high extraction accuracy, using multi-backend extraction (Textract, SageMaker, Bedrock), customizable business rules, and human-in-the-loop verification
- Built end-to-end ML platform capabilities on Amorphic, integrating SageMaker Notebooks and Studio for experimentation, a model registry for managing trained models, and ML inference pipeline nodes for batch predictions
- Designed and deployed **15+ REST APIs** across platform services covering ML model serving, data pipeline execution, and document processing workflows, with request validation, error handling, and response formatting

MapMyIndia

Remote

Software Development Intern

Aug. 2020 – Dec. 2020

- Developed *GeoPark* - a full stack web application for real-time geolocation-based parking availability using React and Node.js

PROJECTS

Ship of Theseus: Authorial Identity Decay | [GitHub](#) | Python, BERT, SBERT, NLP, Statistical Analysis

- Investigated how iterative LLM paraphrasing erodes stylistic markers while preserving semantics, analyzing 66K rows across 7 datasets and 4 paraphraser (ChatGPT, PaLM2, Pegasus, Dipper)
- Built a multi-modal decay analysis pipeline combining lexical (TTR, n-gram overlap), structural (POS, dependency depth), and semantic features (BERTScore, SBERT) to identify the point of no return in authorial identity, with t-SNE decay visualizations

Text-to-SQL RL Optimization | [GitHub](#) | Python, PyTorch, TRL, DSPy, Llama 3.1 8B

- Training Text-to-SQL agents on Spider using GRPO reinforcement learning with execution-based rewards, comparing against SFT and DSPy prompt optimization to measure whether weight-level and prompt-level gains stack
- Designed composite reward function combining execution accuracy, syntax validation, and schema coverage; fine-tuning Llama 3.1 8B via LoRA with Dr. GRPO length-bias correction for stable policy optimization

Chorus: Multi-Agent RAG System | [GitHub](#) | Python, LangGraph, AWS Bedrock, Qdrant, Neo4j, Redis

- Architected hierarchical multi-agent system with 8 specialized agents orchestrated via LangGraph, implementing a query rewrite loop that grades retrieval quality and iteratively refines queries to achieve 90% precision on technical documentation
- Built hybrid search combining vector similarity (Qdrant), BM25 keyword matching, and GraphRAG (Neo4j) with BGE cross-encoder reranking, supporting WebSocket streaming and Redis-backed conversation memory

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, TypeScript, C++

ML & AI: PyTorch, TensorFlow, LangChain, LangGraph, NLP, Scikit-learn, Pandas, NumPy

Cloud & Tools: AWS (Bedrock, SageMaker, Glue, Lambda, EC2, SQS, ALB), Docker, Git, Linux

Databases: PostgreSQL, Aurora, Qdrant, Neo4j, MongoDB, Redis